Concept rendering

SUPERWOOD

SUPERWOOD for Data Centers

Investor companion to Super Mills America · September 2026

Data centers turn SUPERWOOD from a premium skin into a *structural* material

01

The buyer is short of steel and already building with wood

Data centers are among the fastest-growing buyers of structural steel. Customers report backlogs and shortages, and Microsoft and Meta already build data-center structures in mass timber.

02

We turbocharge the wood they already use

Mass timber replaces concrete floors. SUPERWOOD, stronger than A36 steel in tension at one-sixth the weight, adds the steel: members, skins and screens — and strengthens the mass timber itself. It ships today from SuperMill One; truss design and mass-timber enhancement are under way.

03

One basis-of-design win is SuperMill Two-scale demand

A gigawatt campus's skins alone are one to three years of SuperMill One's output; its structure is a year or more of SuperMill Two. The long game is prefabricated envelopes and, eventually, foundations.

The company, team, mills, cost roadmap and the raise are in Super Mills America. This deck covers one application. Strength: company test data vs ASTM A36; campus sizing: company estimate (slides 5–6).

THE BUYER

Data centers are a fast-growing buyer of structural steel, and they are short of it

Concept rendering

500–1,000 t of structural steel per 10 MW

and 5,000–10,000 m³ of concrete — literature intensities for hyperscale builds. A gigawatt campus carries roughly 50–100 kt of structural steel above the slab.

Customers report backlogs and a shortage of structural steel

Supply chain and timeline are the first concern data-center customers raise with us. A domestic material that is lighter to ship and faster to erect answers it directly.

Campuses are built to a standard basis of design

Hyperscalers repeat one design campus after campus. A material written into that design is specified again and again.

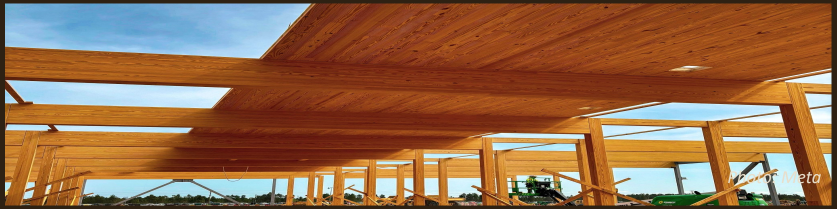
Steel and concrete intensities: arXiv 2509.21312 (Sep 2025), citing Hasan et al. 2022 and Sharma et al. 2023 — secondary literature figures, ranges as published; per-GW figure is that range × 1,000 MW. Shortage and backlog statements are what data-center customers report to InventWood (2026), not a published statistic.

Hyperscalers are already building structures with wood. We *turbocharge* wood.

Microsoft

Two Northern Virginia datacenters with cross-laminated timber floors on a steel frame — about 35% less embodied carbon than conventional steel construction, 65% less than precast. Gensler; Thornton Tomasetti.

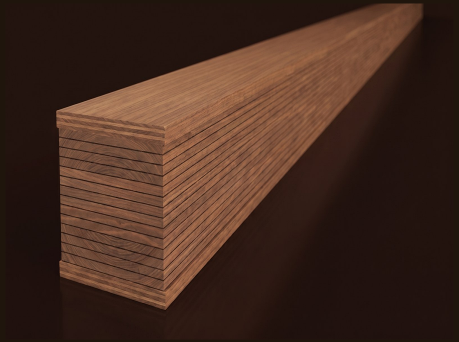
Microsoft Source, Nov 2024



Meta

Mass-timber data-center administrative buildings: Aiken, South Carolina completed 2025 (DPR, SmartLam); Cheyenne and Montgomery under way; about 41% less embodied carbon in the materials substituted.

Meta Sustainability, 31 Jul 2025



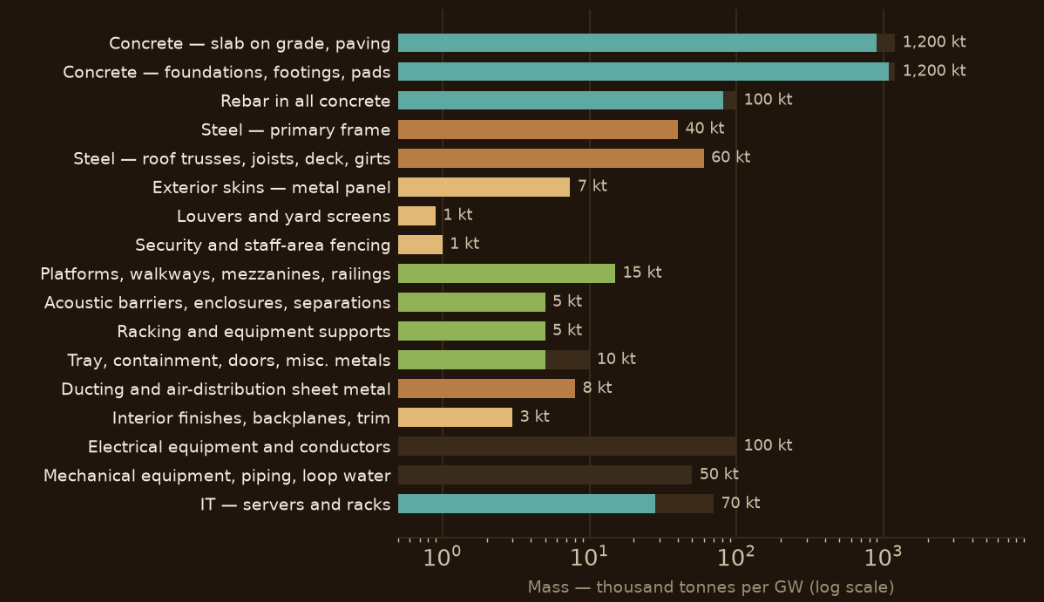
	MASS TIMBER TODAY (CLT, GLULAM)	WITH SUPERWOOD
What it replaces	Concrete floor slabs and decks; some columns and beams	Adds the steel: members, skins, screens and fences — and, ahead, enclosures and foundations
Strength	Lumber-grade; large sections carry the load	Tensile strength above ASTM A36 steel in samples, at one-sixth the weight; thin, dense members
The timber itself	Glulam and CLT at lumber stiffness	Mass-timber enhancement: SUPERWOOD outer laminations (~10% of section) raise a glulam beam's stiffness ~75% and strength ~100%
Exterior use	Needs protection from weather	Exterior grade; resists moisture, pests and rot
Fire	Chars; established assemblies	Chars; far better than ordinary wood — Class A demonstrated in testing
Code path	Established in the building codes	Certifiable under wood standards today; SUPERWOOD-specific standards are the goal

Sources: news.microsoft.com, Nov 2024; Thornton Tomasetti project page; sustainability.atmeta.com, 31 Jul 2025. Logos and photographs are the companies' own, from the cited publications, used to identify the published projects. SUPERWOOD strength: company test data, parallel-to-grain tension. Hybrid-beam gains: derived from beam theory with SUPERWOOD modulus and strength, engineering write-up pending. Beams: concept renderings.

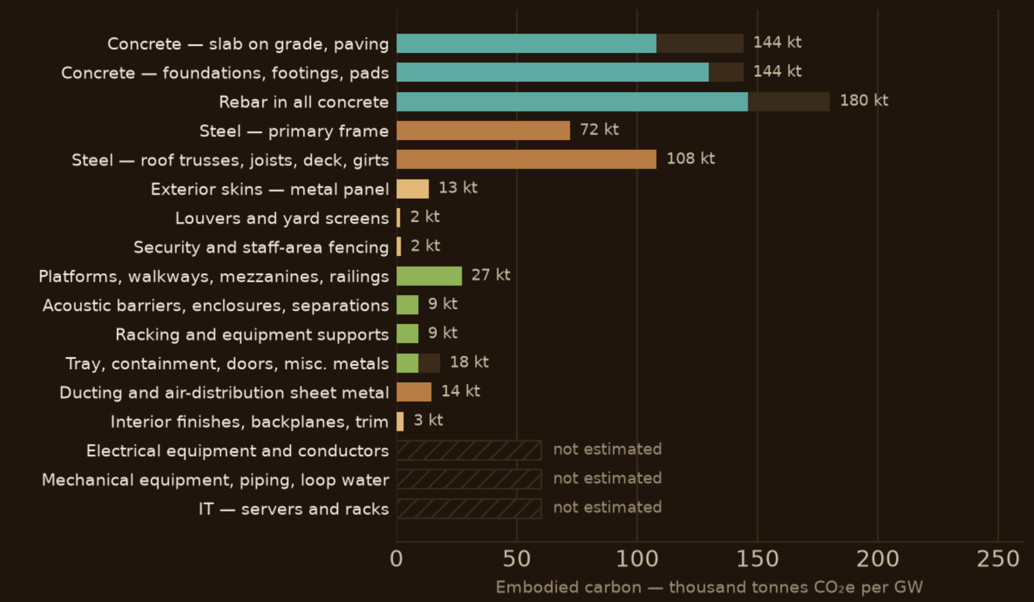
THE SIZE

What a GW data center is made of

BY MASS



BY EMBODIED CARBON



Now — skins, screens, fences, interiors | Next — racks, platforms, barriers, doors | Structural — frame, roofs, enclosures | Long-term vision — slab, foundations, server boxes | Not replaced

50–80% of the above-ground portion of a data center is steel

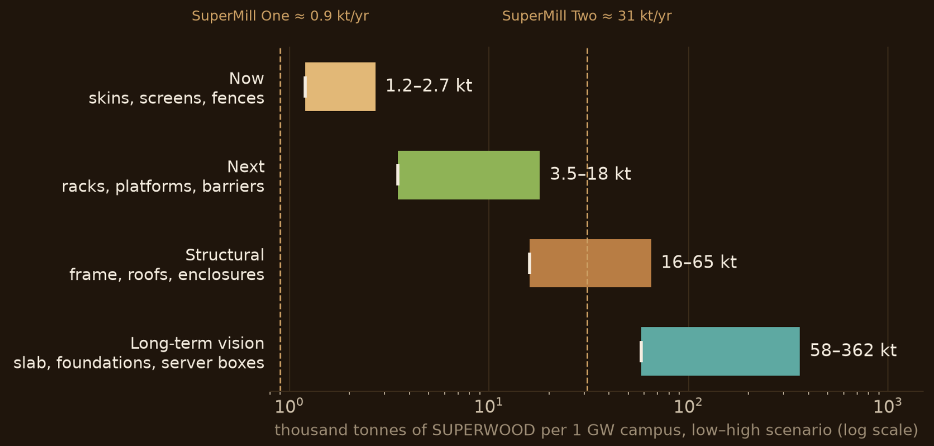
Structure, envelope and contents — frame, roof, skins, platforms, racks, enclosures, equipment. Skins, screens and fences now; racks, platforms and barriers next (racks are a few months of development away); frame and roofs as structure; server boxes eventually — the electronics never.

By embodied carbon, steel is about 60% of the building materials

At the global-average steel factor; about a third with recycled steel. The steel above the slab alone carries roughly 37% of the building materials' embodied carbon — the share SUPERWOOD addresses through the structural horizon.

Company estimate for a 1 GW IT-load campus, high case. Only the concrete and structural-steel intensities are published (arXiv 2509.21312, secondary, conf M); other rows are estimates from unit masses. Carbon factors: steel 1.8 kg CO_2e/kg (global average; recycled 0.4–0.7), concrete 0.12; equipment embodied carbon not estimated. Steel share: company estimate. Model: analyses/materials-mass-and-replacement.xlsx.

What one gigawatt of campus is worth to InventWood



HORIZON	INCUMBENT REPLACED	SUPERWOOD REQUIRED	PLANT-YEARS
Now — skins, screens, fences	3.3–12 kt	1.2–2.7 kt (1.4–3.1M sf)	**1.4–3.1 yr of SuperMill One**
Next — racks, platforms, barriers	12–30 kt	3.5–18 kt	**0.1–0.6 yr of SuperMill Two**
Structural — frame, roofs, enclosures	53–108 kt	16–65 kt	**0.5–2.1 yr of SuperMill Two**
Long-term vision — slab, foundations, server boxes	1,002–2,089 kt	58–362 kt	**1.8–12 yr of SuperMill Two**

Two or three gigawatts of campus absorb SuperMill Two for years. That is the offtake argument and the capacity risk in one number — and why a basis-of-design win with one hyperscaler is the demand anchor for the second mill.

Company estimate (slide 5 model; low–high scenarios). Incumbent replaced is steel except the long-term row (concrete, rebar, server enclosures at 40% of IT mass — estimate; electronics excluded). SUPERWOOD required uses a 0.3–0.6 steel substitution factor (estimate). Plant output: SuperMill One ≈ 0.9 kt/yr (1M sf), SuperMill Two ≈ 31 kt/yr (36M sf) at 0.87 kg/sf, 1.3 t/m³. Pricing is set per application.

We unlock the power of *cellulose*

Our patented process transforms ordinary wood into SUPERWOOD — an environmentally benign process using chemistry common to food and pulp processing.

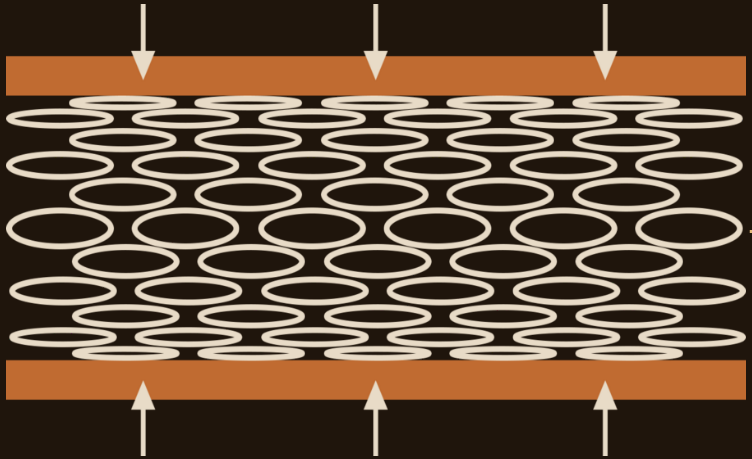
01 · Start with woody feedstock

Hardwoods, softwoods, bamboo, underutilized species, waste wood

Before: open cell vessels

02 · Molecular re-engineering & densification

3–4× density increase, elimination of pores and defects, new hydrogen bonds across fibers



No added glues · No polymer binders

03 · SUPERWOOD

Samples more than 50% stronger than A36 steel in tension, at one-sixth the weight

After: collapsed, interlocking vessels

The *strength* of SUPERWOOD, one number set

500 MPa

tensile strength in production today

600+ MPa

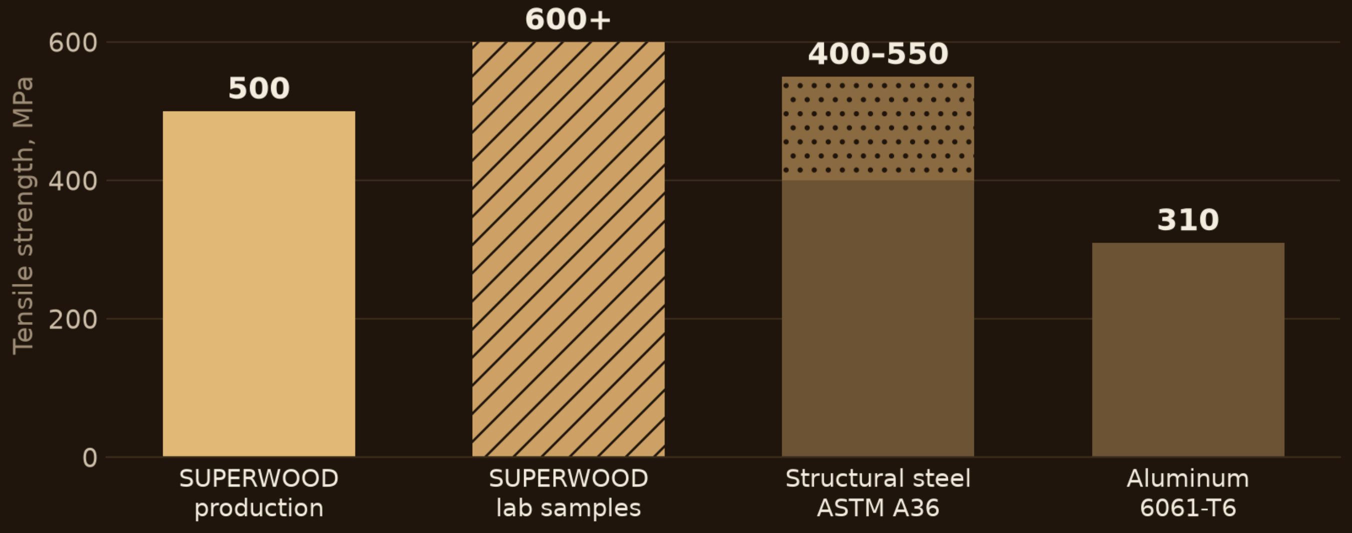
demonstrated in lab samples; pathway toward 1,000

1/6

the weight of steel

7–9×

the strength-to-weight ratio of A36 steel, derived below



HOW THE RATIOS ARE DERIVED

Strength: 500–600 MPa SUPERWOOD ÷ 400 MPa (A36 minimum ultimate tensile) = 1.25–1.5×.

Weight: steel 7.85 t/m³ ÷ SUPERWOOD ~1.3 t/m³ ≈ 6×.

Strength-to-weight: 1.25–1.5 × 6 ≈ 7–9×.

Against the top of the A36 range (550 MPa) the strength ratio is 0.9–1.1×; “stronger than steel” is stated against the A36 minimum.

SUPERWOOD: parallel-to-grain tension, company test data (production QC and lab samples). Steel: ASTM A36 specifies 400–550 MPa ultimate tensile. Aluminum: 6061-T6, 310 MPa typical. Design values and test methods available on request.

Properties that matter on a data center campus

Stronger than steel.
One-sixth the weight.
Made from wood, in America.

FIRE PERFORMANCE
Chars rather than burns — far better in fire than ordinary wood. ASTM E84 Class A demonstrated in testing; specified per order

DURABLE AND IMPACT RESISTANT
Harder and more dent-resistant than oak; impact and storm resistant — suited to fences, screens and door protection

RESISTS MOISTURE, PESTS AND ROT
Exterior grade for facades, yards and fences; no rust; resists termites, mold and fungus

INSULATING, NOT CONDUCTING
Thermal and electrical insulator — no thermal bridging, and non-conductive around electrical infrastructure

SOUND AND VIBRATION DAMPING
Better damping than steel — the basis for acoustic barriers and equipment screens

RF TRANSPARENT
Useful in niches: antenna screening, timing-antenna enclosures, telecom shelters

NATURALLY BEAUTIFUL
The texture, colors and warmth of wood — the biophilic, community-facing surface

Company test data; property data package and test methods available on request. Fire: Class A is a demonstrated capability, not a rating carried by every board.

Speed is the data-center benefit: lighter, *prefabricated*, reconfigurable

01 **One-sixth the weight of steel**

Smaller cranes and crews, easier transport, lighter foundations. Time to power is the metric hyperscalers optimize.

02 **Prefabricated to precision**

Panels and modules built off site and set like mass timber — standardized designs, plug-and-play systems.

03 **Designed for assembly and disassembly**

Reusable, reconfigurable components for infrastructure that changes with every hardware generation.

04 **Mass timber's schedule record**

About 20% average schedule savings across seven case studies, up to 25–30% in others. SUPERWOOD builds the same way.

Concept rendering

Schedule figures are mass-timber case studies (ULI Urban Land: ~20% average across seven case studies, 12.7 vs 15.4 months; other studies 25–30%), used as an analogy for a material that builds the same way. No SUPERWOOD schedule data yet.

THE PATH · NOW

Now: skins, screens and fences from SuperMill One

Every item below ships today as boards up to 8" × 16' × 3/8", exterior or interior grade. SuperMill One makes about one million square feet a year across all markets.

Facades, cladding & rain screens

Biophilic interiors for office space

Louvers & equipment screening

Fencing for staff outdoor spaces

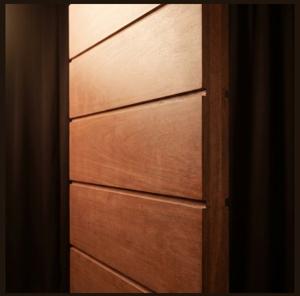
Security fencing around transformers & outdoor infrastructure

Trim, door kicks & sub-framing

Why these first. They need no assembly fire rating — where a finish rating applies, Class A has been demonstrated and is specified per order. They sit where the community and the workforce see the campus. And they are the near-term projects our hyperscaler customers asked for: facades, biophilic office interiors, staff-area fencing, security fencing around critical outdoor infrastructure.

Concept renderings

Next: applications one test program away



Acoustic barrier walls & equipment screens

WHAT IT NEEDS

Acoustic performance data

Walkways & platforms

WHAT IT NEEDS

Load and connection data

Railings

WHAT IT NEEDS

Load testing to code

Window mullions

WHAT IT NEEDS

Thermal and structural data

Interior doors & door protection

WHAT IT NEEDS

Listing for rated openings

Equipment backplanes

WHAT IT NEEDS

Electrical-room listing

Racks & equipment supports

WHAT IT NEEDS

A few months of design and load-data development

Each of these is a scoped test program, not a research question — and the natural first co-funded project with a customer. Which comes first depends on which a customer picks up first.

Concept renderings

Structural: starting *now*, scaling with SuperMill Two

UNDER WAY FROM TODAY'S BOARDS

Structural work has begun. Truss design is starting, and mass-timber enhancement — SUPERWOOD laminations that stiffen and strengthen glulam and CLT-type members — is under way. We expect structural sales on the way to SuperMill Two.

Replacing structural steel is the first thing new data-center customers ask about.

AT SUPERMILL TWO SCALE

Building enclosures & structural components

The co-development target with hyperscaler architects and engineers

Trusses, roofs & long-span members

Roof structure first; fast-to-build enclosures follow

CLT-type floor, roof & wall assemblies

Thin, far stronger panels through the mass-timber product route

Heavy equipment supports & server enclosures

Racks come next, a few months of development; enclosures for servers and equipment follow — the electronics stay

How qualification works. Every structural application is qualified through a scoped test program with the customer who picks it up first, on the pathway mass timber has already opened in the building codes, with insurer acceptance for structural use. Test methods and property data are available on request. This is the Microsoft and Google conversation, and it is SuperMill Two-scale demand.

Concept renderings

Where this goes: prefabricated *envelopes*, then foundations

Concept renderings

Prefabricated building envelopes

Structure and skin shipped as panels and modules and set in days — the co-development target with our hyperscaler customers' architects and engineers. Optimized shells that carry their own loads, weigh a fraction of steel and precast, and can be taken apart and moved.

Foundations, slabs and paving

Lightweight, insulated SUPERWOOD foundations and slabs in place of concrete — installed faster, with lower geotechnical demands, and potentially movable. Concrete is three-quarters of a campus by mass; this is where the material would change the whole building.

Stated technical potential, not an engineered plan: through the structural horizon SUPERWOOD addresses about 5% of campus mass; foundations and slabs would raise the ceiling to about three-quarters — on the order of 1–2 million tonnes of concrete and rebar per gigawatt. No design or code pathway exists yet.

THE EVIDENCE

Every account on one ladder

CONVERSATIONS · APPLICATIONS IDENTIFIED · PROJECTS SCOPED · TESTING & MOCKUPS · FIRST PURCHASE · BASIS OF DESIGN

WHAT THEY ARE DOING WITH US

Microsoft

Next: First skin project on one building or yard

Facades, biophilic interiors, staff and security fencing under discussion; roof-truss and enclosure work with their architecture and engineering firms; interest in SUPERWOOD in the basis of design

Meta

Next: Backplane listing; first facade project

Engaged for over a year across the data-center ecosystem; facade applications and backplanes under way; structural applications and racking to follow

Google

Next: Scope with their design teams

Engagement just begun; initial focus on replacement of structural steel

Vertiv · Wooden Data Center

Next: Identify first applications

Equipment and modular data-center builders exploring SUPERWOOD components

Data center operators

Next: Fence specification and quote

Operators focused on security fencing around outdoor infrastructure

ENGINEERING AND CONSTRUCTION PARTNERS Fast + Epp has run over a thousand small-scale experiments on SUPERWOOD, funded by the Canadian government · Timber Engineering (structural engineering) · HITT and Turner, contractors, are advocates — they build, they do not specify

Engagement stages and partner statements as reported by InventWood, September 2026; next steps undated. Public-record backup for Microsoft and Meta on the next slide.

Microsoft and Meta: on the record, and with us

Microsoft *broad engagement*

ON THE PUBLIC RECORD

Two Northern Virginia datacenters built with cross-laminated timber floor panels on a steel frame — an estimated 35% embodied-carbon reduction vs. conventional steel construction and 65% vs. precast concrete. Design by Gensler; structural engineering by Thornton Tomasetti.

Microsoft Source, Nov 2024; Thornton Tomasetti project page

WORKING WITH INVENTWOOD

- **Strategic interest**
  Incorporating SUPERWOOD into the basis of design for future data centers
- **Application areas**
  Roof trusses, leading to roofs and fast-to-build enclosures — with their principal architecture and engineering firms
- **Immediate opportunities**
  Building facades · biophilic interiors · staff-area fencing · security fencing around transformers

Concept rendering — security fencing around a transformer yard

Meta *engaged for over a year*

ON THE PUBLIC RECORD

Piloting mass timber for data center administrative buildings: first completed in 2025 at Aiken, South Carolina (DPR, SmartLam); projects under way in Cheyenne, Wyoming and Montgomery, Alabama — about a 41% reduction in the embodied carbon of the materials substituted.

Meta Sustainability, 31 Jul 2025

WORKING WITH INVENTWOOD

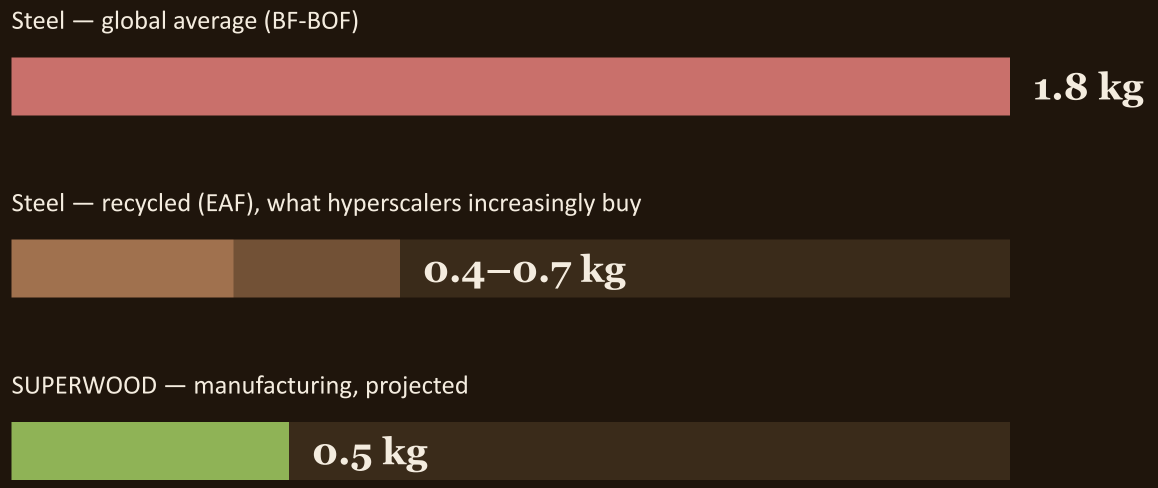
- **Broad engagement**
  Across every part of the data-center ecosystem, for more than a year
- **Under way**
  Facade applications and equipment backplanes
- **Next**
  Structural applications and racking over time

Concept rendering — equipment backplane in an electrical room

Engagement statements as reported by InventWood, September 2026. Sources: news.microsoft.com, Nov 2024; thorntontomasetti.com; sustainability.atmeta.com, 31 Jul 2025.

The carbon claim, sized and labeled: a projection until the LCA lands

KG CO_2E PER KG OF MATERIAL, CRADLE TO GATE



Biogenic carbon, reported separately. SUPERWOOD stores about 1.3 kg CO_2e of biogenic carbon per kg. Under EN 15804 it is released again in the end-of-life module, so it nets to zero over the life cycle. The durable advantage is manufacturing emissions, not storage.

PER FUNCTIONAL UNIT — REPLACING ONE TONNE OF STEEL

SUPERWOOD needed: 0.3–0.6 t per tonne of steel replaced (substitution factor, estimate — to be engineering-stamped per element).

SUPERWOOD manufacturing emissions: 0.3–0.6 t × 0.5 = 0.15–0.30 t CO_2e.

Against average steel (1.8 t): a reduction of 83–92%.

Against recycled steel (0.4–0.7 t): a reduction of 25–79%.

Any stated reduction must name its steel baseline and its substitution factor. Both are shown here; neither is yet verified.

Pre-LCA projection for a full-scale plant. LCA under way with Prof. Ming Hu, University of Notre Dame.

SUPERWOOD 0.5 kg/kg manufactured and 1.3 kg/kg biogenic: company projections, pre-LCA. Steel 1.8 kg/kg: global BF-BOF average. EAF 0.4–0.7 kg/kg: typical published range for recycled-scrap steel [conf: M]. Substitution factor 0.3–0.6: company estimate. Reductions are arithmetic on these inputs.

What has to be true, and what could stop it

RISK	HOW WE HANDLE IT
Insurers and code officials must accept SUPERWOOD for structural use	Start with skins and non-structural items that need no assembly rating. Qualify structural applications with the first customer, on the pathway mass timber opened.
Qualification could run longer than customer design cycles	Sell what needs no qualification now. Hyperscalers design campuses years ahead, so basis-of-design work starts before qualification ends.
One plant, shared across every market	SuperMill One allocates output; a gigawatt's skins alone are one to three plant-years. SuperMill Two is the answer — and the raise.
Price against metal panel, fiber cement, CLT and recycled steel	Premium skins where appearance and community acceptance carry value. Structural competitiveness arrives with SuperMill Two cost (roadmap basis, not yet realized).
Carbon figures are projections until the LCA lands	Labeled pre-LCA everywhere; LCA under way with Prof. Ming Hu, University of Notre Dame. No carbon claim without its baseline and substitution factor.
Customer concentration	Three hyperscalers plus Vertiv, Wooden Data Center and operators — and what is qualified for a data center is qualified for the wider structural market.

No probabilities are assigned; these are the conditions the data-center thesis depends on, stated so they can be tracked.

A concrete path to first projects

1 **Skins on one building or yard**

Facades, interiors, louvers, staff-area or security fencing — installed from SuperMill One output on a friendly site. Produces an installed reference and installed-cost data.

2 **A scoped test program with that customer**

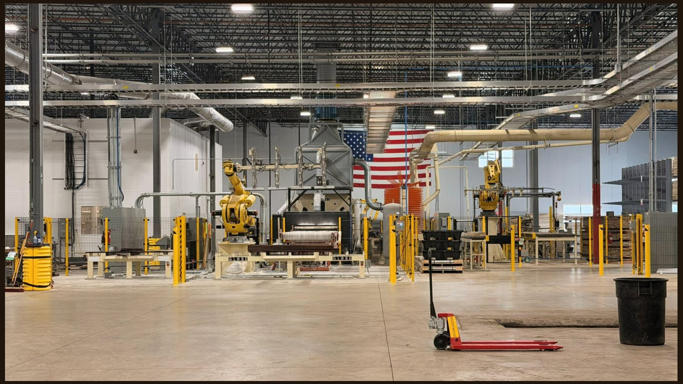
Toward whatever the chosen application needs, co-funded. Produces the data package the next application inherits.

3 **Basis-of-design work with the customer's architect and engineer**

Structural enclosure and truss solutions that can be written into the standard campus design. Produces a specification that repeats.

4 **Measure the carbon delta**

A baseline bill-of-materials comparison and verification plan. Produces defensible numbers as the LCA completes.



SuperMill One, Frederick, MD

Concept for SuperMill Two

Steps are sequenced, not dated. Each names what it produces.

Let's build what's *next*.

SUPERWOOD

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